

An Analysis and Simulation of Defending Tracking Control Neural Network for Known Affine Discrete-Time Nonlinear Systems Against Adversarial Attacks

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1 Introduction

1.0.1 Utilizing Close Analogs to Understand The Difference Equation with Variable Structure

The affine in the inputs discrete-time nonlinear dynamic system will be considered. Both a semantic and mathematical description of the system will be provided.

Definition 1. *Affine-In-The-Inputs Discrete-Time Nonlinear System_[1]*

$$\underbrace{x(k+1)}_{\substack{\text{next} \\ \text{system-state}}} = \underbrace{x(k)}_{\substack{\text{current} \\ \text{system-state}}} + \underbrace{\tau}_{\substack{\text{discrete-time} \\ \text{step}}} \left[\underbrace{f(x(k))}_{\substack{\text{state} \\ \text{dynamics}}} + \underbrace{g(x(k))}_{\substack{\text{coupling} \\ \text{dynamics}}} \underbrace{u(k)}_{\substack{\text{control} \\ \text{input}}} \right] \quad (1)$$